

# HSIU-CHI LU

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## Research Interests

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Computational Social Science, Network Science, Relational Sociology, Complex Systems, Polarization, Information Diffusion

## Education

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**Northeastern University, Network Science Institute**, Boston, USA Sep. 2025 –  
Doctor of Philosophy in **Network Science**  
Advisor: David Lazer

**National Chengchi University**, Taipei, Taiwan Aug. 2022 – Aug. 2024  
Master of Science in **Sociology**  
Advisor: Hsuan-Wei Lee, Jen-Hao Chen  
Thesis: *Agents of Discord: Modeling the Impact of Social Bots on Opinion Polarization in Complex Networks*

**National Chengchi University**, Taipei, Taiwan Aug. 2018 – Jul. 2022  
Bachelor of Science in **Sociology**  
Minor in **Applied Mathematics**

## Research Papers

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### Peer-Reviewed Journal Articles

- **Hsiu-Chi Lu** & Hsuan-Wei Lee (2024). “Agents of Discord: Modeling the Impact of Political Bots on Opinion Polarization in Social Networks.” *Social Science Computer Review*. Advance online publication. [Link](#).  
\*Presented at **Sunbelt 2024**, International Network for Social Network Analysis (INSNA), Edinburgh, Scotland, Jun. 2024.
- **Hsiu-Chi Lu** & Hsuan-Wei Lee (Forthcoming). “Modeling Disinformation Resilience: How Social Pressure and Network Dynamics Drive and Sustain the Spread of False Narratives.” *Humanities & Social Sciences Communications*. (Accepted).  
\*Presented as a poster at **IC2S2 2024**, Philadelphia, PA, Jul. 2024. [Link](#).

### Manuscripts Under Review and Revision

- **Hsiu-Chi Lu** & Jen-Hao Chen (equal contribution). “A Journey of Inclusiveness: Mapping Emotions and Phrases for LGBTQ in Major Taiwanese News Outlets over the Past Decade.” *Revise and Resubmit (Major Revision) at Journal of Homosexuality*.  
\*Presented as a poster at **IC2S2 2024**, Philadelphia, PA, Jul. 2024. [Link](#).

### Selected Conference Presentations

- **Hsiu-Chi Lu** (2021). “Analysis and Reflection on the Relationship and Actions Between AI and Humans in the Information Age: A Case Study of GPT-3 and its Creations.” **2021 Annual Meeting of the Taiwanese Sociological Association**, Taipei, Taiwan, Nov. 2021. (Presented in Mandarin). [Link](#).  
\*Selected Undergraduate Research Project funded by the National Science and Technology Council (NSTC).

## Academic Experience

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**Research Assistant** Aug. 2025 – Present  
Network Science Institute, Northeastern University Boston, MA, U.S.A.

- Conducted independent computational social science research utilizing over a decade of large-scale panel data (including U.S. X users) to investigate cultural landscapes, lifestyle polarization, and digital data biases.
- Processed, analyzed, and visualized complex datasets from the National Internet Observatory (NIO), synthesizing empirical findings into comprehensive technical reports.

**Research Assistant** Aug. 2024 – Aug. 2025  
College of Health, Lehigh University Bethlehem, PA, U.S.A.

- Developed a model to investigate disinformation spread in online social networks, focusing on mechanisms that lead to widespread dissemination; resulting in a paper currently under review.
- Implemented mathematical functions to model experiential data on disinformation, identifying and defining underlying dissemination patterns.

**Research Assistant** Feb. 2022 – Aug. 2025  
Department of Sociology, National Chengchi University Taipei, Taiwan

- Conducted independent and collaborative research on LGBTQ representation in major Taiwanese news outlets during the 2010s, resulting in a conference presentation and a paper currently under review.
- Developed a structured database of approximately 20,000 records by utilizing Python-based web crawling to collect over 50,000 news articles from major Taiwanese media outlets. Employed natural language processing, machine learning, and large language models to filter and classify data.
- Applied statistical methods to analyze patterns between sexual orientation and external factors, investigating societal inclusiveness and enhancing clarity through data visualization.
- Analyzed night work patterns and their relationship with marital status among Taiwanese individuals, contributing insights to a work-life balance report; additionally, used Airtest to monitor mobile devices and gather data on college students' mental health.

#### Research Assistant

Nov. 2022 – Jul. 2024

Institute of Sociology, Academia Sinica

Taipei, Taiwan

- Conducted research on digital society issues such as misinformation, online polarization, and political bots, resulting in one published paper.
- Designed and visualized agent-based simulations in Python to examine the interaction between political bots, polarization, and the spread of disinformation.
- Analyzed network data to explore the role of network structure in opinion diffusion and disinformation dissemination.

#### Lab Intern

Jul. 2023 – Dec. 2023

Institute of Information Science, Academia Sinica

Taipei, Taiwan

Lab: **Natural Language Processing and Sentiment Analysis Lab**

- Supported the laboratory team in crafting a questionnaire for experiments on cognition of misinformation and processing demographic data from respondents.
- Contributed to interdisciplinary collaborations by integrating social science expertise into AI research endeavors.

#### Data Scientist

Feb. 2022 – Jun. 2023

Big Data Group, Taiwan Institute of Governance and Communication Research

Taipei, Taiwan

- Developed Python-based web-crawling scripts to gather social media data, creating three databases from Facebook and Ptt (a Taiwanese Reddit-like forum). The codes are available on [here](#), and the datasets can be requested via this [link](#).
- Structured and analyzed social media data using natural language processing and statistical methods to explore public opinion on the pandemic and the 2022 local elections in Taiwan.

### Teaching Experience

#### Teaching Assistant

Sep. 2022 – Feb. 2023

National Chengchi University.

Course: **Data Analysis in Sociology**

- Prepared and conducted a one-hour discussion class on R and statistics each week.
- Supported over 70 students in group settings with their final papers, assisting them in discussing research questions, solving statistical problems, and providing programming support.

### Awards and Grants

- **Master Dissertation Award.** Taiwanese Sociological Association 2024
- **Travel Grants.** 10th International Conference on Computational Social Science 2024
- **Outstanding Student Overseas Conference Grant.** Foundation for the Advancement of Outstanding Scholarship 2024
- **Subsidy for Students Attending International Conference.** Ministry of Science and Technology, Taiwan 2024
- **Undergraduate Research Project Scholarship.** Ministry of Science and Technology, Taiwan 2020  
Research Topic: Analysis and Reflection on the Relationship and Actions Between AI and Humans in the Information Age: A Case Study of GPT-3 and its Creations
- **Presidential Award,** National ChengChi University 2019

### Research Skills

- **Network Science Analysis**
- **Web Crawling**
- **Machine Learning**
- **Natural Language Processing**
- **Quantitative and Qualitative Methods**
- **Agent-Based Simulation**
- **Data Processing and Visualization**
- **Programming:** Python, R, Latex